

Project 6 - Modular Terraform Project

Project 6: Modular Terraform Project

Objective:

Build reusable Terraform infrastructure using modules for Azure resources.

Project Structure:

terraform-azure-modules/

 modules/

 network/

 vm/

 environments/

 dev/

 prod/

 README.md

Key Concepts Learned:

- Terraform Modules
- Reusable Infrastructure
- Environment Separation
- Clean Project Structure
- GitHub Version Control
- Azure VM Deployment

Modules Created:

1. Network Module

- Virtual Network
- Subnet
- Network Security Group

2. VM Module

- Linux Virtual Machine
- Public IP
- Network Interface
- SSH Authentication

Environments:

- Dev Environment
- Prod Environment

GitHub Issue Resolved:

Removed large Terraform provider binaries and .terraform cache files from Git history.

Important Files Ignored:

- .terraform/
- terraform.tfstate
- terraform.tfstate.backup
- provider binaries

Skills Gained:

- Terraform module development
- Infrastructure organization
- Multi-environment deployments
- Git troubleshooting
- Azure infrastructure automation